

Supplementary information for A traceable time-series power-flow dataset for Portugal's high-voltage grid

This document supplements the Data Descriptor. Machine-readable field definitions, device-level audits, complete contingency membership, file hashes, and replay commands are provided in the versioned release data dictionary, CSV/JSON audit products, and release manifest.

Table S1 | Mapping from source fields to standardized semantics.

Source / source field	Standard meaning or target field	Transformation	Join and missing-data rule
E-REDES energia	Nodal p_mw	15-min kWh / 250 to MW	Match codigo_subestacao to facility; no interpolation
E-REDES datahora ; data / hora	UTC interval start	Subtract 15 min from end label; Europe/Lisbon to UTC	Average duplicate labels by station and flag ambiguity
REN source_date + source_index	calendar.timestamp_utc	Construct from Lisbon civil date and within-day index	Index distinguishes repeated daylight-saving hour
REN series_name + value_mw	National total by energy, load, or storage type	Preserve MW; allocate within type	Missing required series fails the case
RND line ID / tensao_de	source_line_id / voltage_kv	Preserve source key; standardize voltage to kV	Restrict to voltage levels in this release
Source geometry / coordinates	Geospatial geometry; longitude / latitude	Store in WGS 84; compute distances in EPSG:3763	Invalid geometry is recorded
OSM way / relation / power tags	Source ID, circuit identity, asset type	Split only at explicit nodes; retain relation keys	Geometric crossing does not imply connection
Public nameplate capacity and rating	nameplate_mw / sn_mva	Standardize power and capacity units	Missing public value uses a flagged proxy
Archived URL / bytes / hash	provenance.raw_files	SHA-256 and relative archive path	Retain source-file locator

The table describes semantic mappings and does not imply identical raw field names in every source file. The archived source files remain authoritative for exact raw fields.

Table S2 | Default electrical-parameter library.

Voltage (kV)	r (Ω /km)	x (Ω /km)	c (nF/km)	Imax (kA)	Status
60	0.1093	0.368854	9.2	0.606	Voltage-class engineering proxy
130	0.07	0.33	9.8	0.8	Voltage-class engineering proxy

Voltage (kV)	r (Ω/km)	x (Ω/km)	c (nF/km)	Imax (kA)	Status
150	0.06	0.32	10.0	0.85	Voltage-class engineering proxy
220	0.04	0.285	11.0	1.2	Voltage-class engineering proxy
400	0.025	0.25	12.0	2.0	Voltage-class engineering proxy

These are summer baseline defaults rather than final values for every line. Source-backed values override individual fields. Cable factors are $r \times 0.65$, $x \times 0.35$, $c \times 20$, and $\text{Imax} \times 0.90$; seasonal rating factors are given in Table 6 of the main manuscript. Values are frozen in `model_config.json`.

Transformer voltage pair (kV)	Default S (MVA)	vk (%)	vk _r (%)
400/220	450.0	12.0	0.3
400/150	450.0	12.0	0.3
400/60	170.0	12.0	0.35
220/150	250.0	12.0	0.35
150/130	140.0	12.0	0.45
220/60	170.0	12.0	0.4
150/60	170.0	12.0	0.45
130/60	126.0	12.0	0.45

Default transformer capacities and impedances are engineering proxies. Published capacities take precedence. When aggregate capacity for a voltage pair is lower than the REN inventory, capacities are scaled upward and the factor is retained. Device-specific overrides, including Estoi, take precedence over these defaults.

Table S3 | Principal mapping and evidence fields.

Storage location	Principal fields	Interpretation
<code>grid.generators</code>	<code>source_id</code> ; <code>bus_id</code> ; <code>bus_assignment_rule</code> ; <code>available_from_utc</code>	Asset source, bus mapping, and commissioning treatment
<code>grid.buses</code>	<code>endpoint_match_distance_m</code> ; <code>source_status</code>	Source-backed versus derived buses
<code>grid.lines</code>	<code>parameter_status</code> ; <code>r/x/c/max_i_status</code>	Field-level parameter evidence
<code>scenario.load_operating_points</code>	<code>observed_p_mw</code> ; <code>ren_residual_p_mw</code> ; <code>q_mvar</code>	Observed load and spatially allocated residual
<code>monthly_model.audit_bundles</code>	Allocation, boundary, and loss records	Case-level conservation audit
<code>provenance.raw_record_locator</code>	<code>entity_id</code> ; <code>raw_table</code> ; <code>raw_record_id</code>	Link from derived entity to source record

Table S4 | Quality controls and failure handling.

Layer / check	Threshold or condition	Action	Record location
Static / identifiers and endpoints	Valid equipment key and existing endpoints	Block invalid connection	Static audit and provenance

Layer / check	Threshold or condition	Action	Record location
Static / voltage and length	Same-voltage connection, length >0, no self-loop	Block invalid branch	Topology audit
Static / connectivity	Components, isolated nodes, boundary inventory	Record and interpret within scope	<code>topology_summary.json</code>
Time series / required inputs	Consistent UTC keys and required series present	Retain error; do not interpolate	<code>monthly_model.cases.error</code>
Time series / load conservation	Nodal sum differs from Consumption + Storage by 0.11 MW	Block case if failed	Allocation audit
Time series / generation capacity	Commissioned mapped assets satisfy 0 P nameplate	Excess becomes an explicit national proxy	Generation-allocation audit
Solution / AC convergence	Newton–Raphson tolerance 10^{-6} MVA; 50 iterations	Retain failed status	<code>monthly_model.cases</code>
Solution / active-power closure	Record Pgen + Pnet – Pload – Ploss	Report numerical residual	Audit bundle and validation summary
Screening / voltage and thermal	0.90–1.10 p.u.; normally 100% rating	Flag; retain completed case	Cases and state arrays
External comparison	Common time or region; pairwise missing exclusion	Retain anomalous months and report n	Comparison output and confidence intervals
Database write	Summary, arrays, and audit bundle in one transaction	Roll back full case and retain error	Monthly transaction log
Release integrity	Expected, complete, array, and audit counts agree	Do not mark month complete if failed	Run manifest and hashes

Contingency thresholds and exclusions are defined in Table S8; routine case-quality checks and contingency screening are distinct.

Supplementary Notes

Supplementary Note S1 | Data layers and joins

Table S5 | Data layers, grain, and principal join keys.

Data layer	Grain	Principal join keys	Purpose
<code>provenance.raw_files / raw_*</code>	File / source record	<code>artifact_id</code> ; <code>raw_record_id</code>	Source evidence and fingerprints
<code>grid.*</code>	Equipment	<code>model_id</code> ; equipment ID; <code>bus_id</code>	Static network and asset mapping
<code>geo.*</code>	Geometry	Equipment or facility ID	Spatial representation
<code>main.interval_calendar / ren_dispatch / eredes_load</code>	15-min interval or raw record	<code>timestamp_utc</code> ; source time key	Common time-series inputs
Scenario operating-point tables	Time × equipment	<code>model_id</code> ; equipment ID; time key	Nodal inputs and boundary
<code>monthly_model.cases / state_arrays</code>	Case / case array	<code>case_id</code> ; <code>timestamp_utc</code>	State summary and equipment results
Monthly audit bundles / <code>run_manifest</code>	Case / monthly run	<code>case_id</code> ; <code>run_id</code>	Allocation audit and integrity
<code>raw_eredes_aux / scenario.annual_nminus1_*</code>	Validation or contingency case	Region, facility, or time × element key	External validation and examples

Supplementary Note S2 | Release and model-version reconciliation

The unique frozen snapshot is **SimPT60-2026.09.21-r1**. CORE-3783 fixes the static network used for the 31,492 historical cases and the main Technical Validation: 3,783 buses, 4,943 line rows, and 228 transformer rows. N1-3787 is restricted to the contingency examples in Supplementary Notes S4–S5.

N1-3787 adds four colocated connection nodes to split the Lanheses–Feitosa topology representation and reconnects the endpoints of five lines without changing the line-row count. It retains 228 transformer rows. In the Estoi row, the **parallel** field changes from one to three and per-unit **sn_mva** from 170 to 126 MVA, so the model represents 230 equivalent transformer units while preserving 228 transformer rows and 228 transformer outage groups. An Estoi unit outage decrements **parallel** from three to two. The representation does not establish that all three physical units were simultaneously in service at every study time.

The differences were not written back to the historical monthly databases. Users must select a named model variant for each analysis. Field-level differences, static and asset SHA-256 hashes, the locked environment, minimal replay command, and 22 regression snapshots are stored in **release.json**, **SHA256SUMS**, the release README, and **model_field_diff.csv**.

Supplementary Note S3 | Additional validation and sensitivity

Parameter and spatial-allocation experiments use 22 representative times. Table S6 gives paired statistics; Figure 5 in the main manuscript shows the rank, Top-20 set, and maximum-loading responses.

Table S6 | Parameter and spatial-allocation sensitivity.

Configuration	Converged / cases	Median / minimum	Median / minimum Jaccard	Maximum absolute Δ loading (pp)	Median absolute Δ loss rate (pp)
Shared baseline	22 / 22	1.000 / 1.000	1.000 / 1.000	0.00	0.000
R/X +20%; C –20%	22 / 22	0.9988 / 0.9985	0.905 / 0.739	6.08	0.284
R/X –20%; C +20%	22 / 22	0.9981 / 0.9977	0.905 / 0.739	6.56	0.278
Current rating +20%	22 / 22	1.0000 / 1.0000	1.000 / 1.000	22.98	0.000
Current rating –20%	22 / 22	1.0000 / 1.0000	1.000 / 1.000	34.48	0.000
Transformer S –20%, Z +20%	22 / 22	0.9948 / 0.9931	0.779 / 0.600	17.10	0.087
Transformer S +20%, Z –20%	22 / 22	0.9954 / 0.9950	0.818 / 0.538	13.78	0.058
Uniform boundary weights	22 / 22	0.9971 / 0.9939	0.905 / 0.667	17.37	0.183
Voltage- weighted boundary	22 / 22	0.9967 / 0.9936	1.000 / 0.739	7.81	0.089
Same- technology capacity allocation	22 / 22	0.9937 / 0.9642	1.000 / 0.818	1.49	0.023

Configuration	Converged / cases	Median / minimum	Median / minimum Jaccard	Maximum absolute Δ loading (pp)	Median absolute Δ loss rate (pp)
Residual load by transformer capacity	22 / 22	0.9962 / 0.9947	0.818 / 0.429	6.50	0.096
Residual load uniform by delivery point	22 / 22	0.9963 / 0.9952	0.905 / 0.538	9.91	0.086

Each configuration uses 22 times. After removing the repeated baseline, all 264 cases converged. Loss rate is network loss divided by model load; changes are percentage points.

Hotspot persistence is computed within each time across 12 configurations. Seventy-four of 4,943 lines enter the Top 20 at least once, and 56 reach a within-time frequency of at least 0.8. The highest frequency across all experiments is 207/264=78.4%; no line appears at every time and configuration.

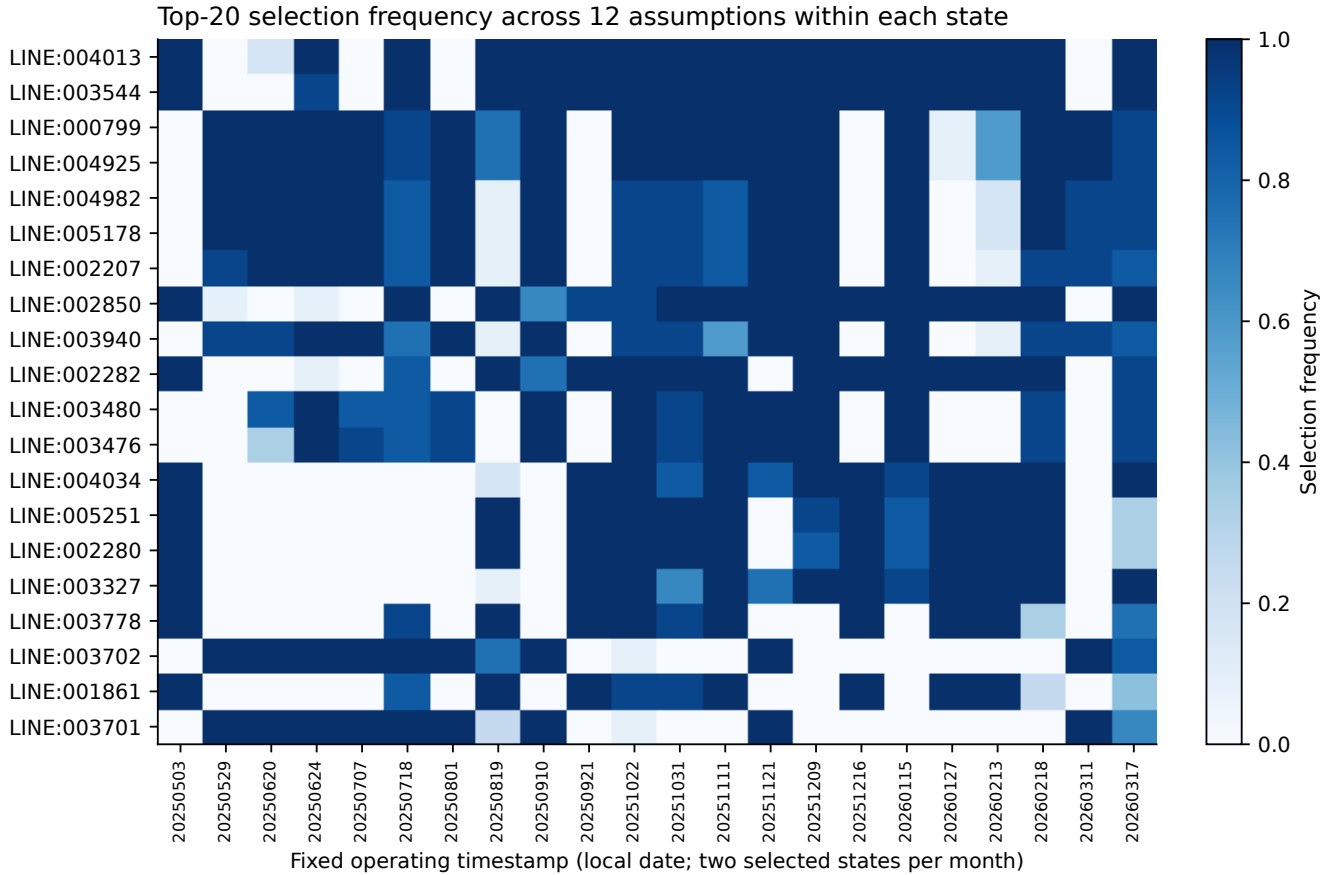


Figure S1 | Hotspot inclusion frequency under alternative assumptions. Each cell gives the proportion of the 12 configurations at one time in which a line enters the Top 20. Complete membership and frequencies are released as CSV.

The station-held-out experiment uses substations as five deterministic folds and compares global capacity share, five nearest geographic stations, and five nearest stations on the reconstructed network. Inverse distance weights are used; network distance uses in-service line lengths and a small positive transformer-edge length. Confidence intervals use a station-cluster bootstrap.

The operating-proxy ablation changes reactive-power limits, PV voltage targets, load compensation, shunt reactors, and tap control at the same 22 times. Table S7 reports the largest paired changes relative to the archived baseline.

Table S7 | Operating-proxy ablation; maximum absolute change over 22 paired times.

Configuration	Converged / attempted	Δ minimum voltage (p.u.)	Δ maximum line loading (pp)	Δ maximum transformer loading (pp)
Baseline	22 / 22	0.00000	0.000	0.000
Legacy fixed neutral tap	22 / 22	0.00000	0.000	0.000
No compensa- tion/reactors; fixed taps	22 / 22	0.00917	0.248	0.892
No load compensation	22 / 22	0.00956	0.182	0.890
Five reactors disabled	22 / 22	0.00090	0.229	0.176
PV target 0.99	22 / 22	0.01120	1.370	0.762
PV target 1.01	22 / 22	0.01117	1.337	0.740
Reactive-power limits $\times 1.5$	22 / 22	0.00975	0.351	0.489
Reactive-power limits $\times 0.5$	22 / 22	0.02845	1.419	0.423
Ratio taps active with voltage control	22 / 22	0.00903	0.032	4.305
Ratio taps active but fixed neutral	22 / 22	0.00000	0.000	0.000

Supplementary Note S4 | Detailed contingency diagnostics

At one stressed snapshot, targeted diagnostics screen 50 highly loaded physical line circuits and 20 in-service transformers. Sixty-seven of 70 cases converge under the specified reactive-power boundary and 66 pass all screens. Three northern line cases yield diagnostic solutions only after relaxing the reactive-power boundary and show thermal violations; one radial circuit creates a material supply-loss island.

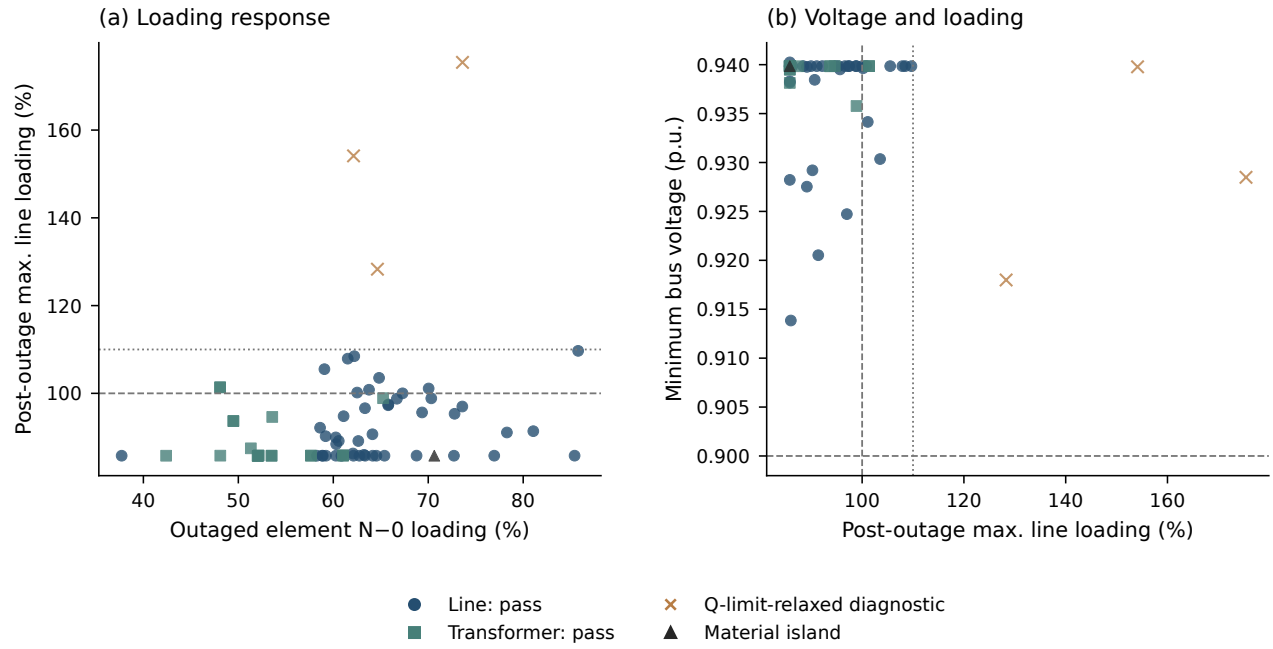


Figure S2 | Targeted AC N-1 response. Circles and squares denote line and transformer cases that pass screening; crosses denote diagnostic-only solutions; triangles denote material islands. Dashed lines show continuous-rating and voltage-screening limits.

The full-element panel uses the same 1,649 outage groups at all six times. Table S8 defines selection and exclusion; `nminus1_exclusions.csv` and the result database contain identifiers, membership hashes, endpoint state, and case-level output.

Table S8 | N-1 outage universe at the representative times.

Level	Count	Rule
Source line rows	4,943	Retain series segments; a row is not necessarily a physical circuit
Excluded line rows	172	156 out of service, 2 station busbars, 14 without finite base result
Eligible line rows	4,771	In service, not a busbar, and finite base result
Line outage groups	1,421	Operate series segments together; remove one circuit from parallel equivalents
Transformer rows / outage groups	228 / 228	One group per row; Estoi's row represents three parallel units
Equivalent transformer units	230	Estoi contributes three units; one unit is removed by a 3→2 parallel decrement
Outages per time	1,649	Identical membership at all six times

“Without finite base result” is distinct from zero current and from out-of-service status; interpretation requires the service flag, connectivity, and solution state together.

The original incremental diagnostic requires principal power-flow convergence, no material island, and no new voltage

or thermal violation relative to $N=0$. It does not detect worsening of a pre-existing line violation. For outage (k), time (t), and the physical-circuit set ($O_0(t)$) that already violates the limit, define

$$W(k, t) = \max_{c \in O_0(t)} \max [0, L_k(c, t) - L_0(c, t)].$$

Table S9 uses one percentage point as a study diagnostic tolerance. This supplements rather than replaces the absolute pass count.

Table S9 | Worsening of pre-existing line violations.

Time	Base violating circuits	Cases with $W > 1$ pp	Maximum W (pp)	Incremental pass but $W > 1$ pp	Incremental pass and $W \geq 1$ pp
Maximum net export	0	0	0.000	0	1,527
Maximum net import	0	0	0.000	0	1,455
Maximum photovoltaic	0	0	0.000	0	1,526
Maximum wind	3	23	114.924	17	1,485
Minimum load	0	0	0.000	0	1,538
Maximum load	1	20	15.023	15	1,501

Supplementary Note S5 | Worked application examples

A deterministic grid-stress example starts from 2026-01-20 19:45 UTC and jointly scales nodal load, generation by technology, and boundary exchange from 0.90 to 1.15. All five cases converge. Maximum line loading is 96.10% at baseline, 100.50% with three overloaded lines at 1.05, and 109.41% at 1.15. Minimum voltage remains above 0.90 p.u. Exact case output is released as CSV.

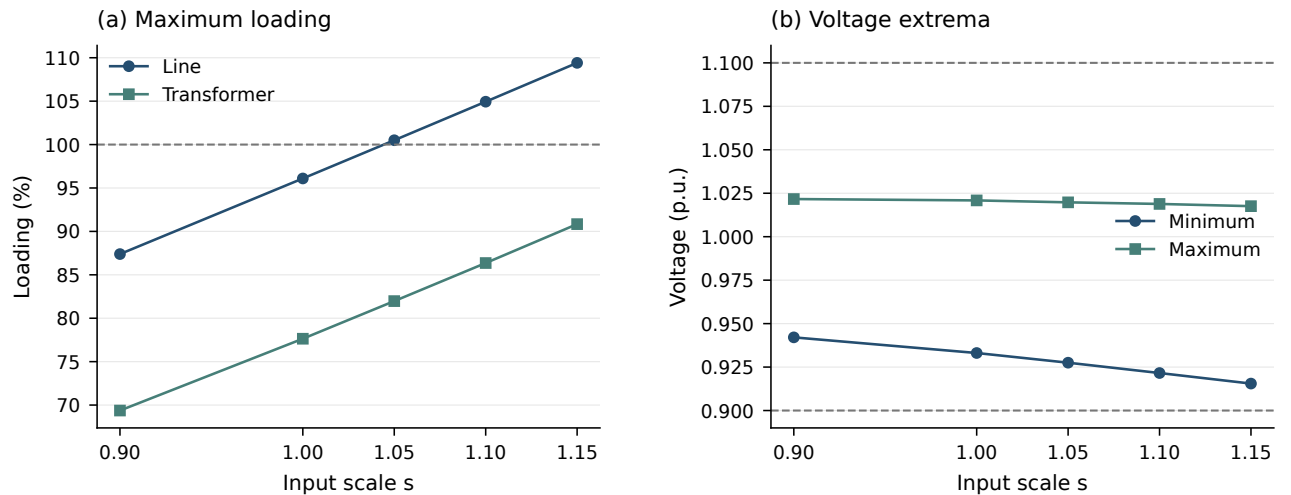


Figure S3 | Deterministic grid-stress response. Maximum line and transformer loading and minimum and maximum bus voltage. Dashed lines mark 100% line loading and the 0.90–1.10 p.u. screening band.

Six times are selected in advance: maximum and minimum load, maximum wind, maximum photovoltaic generation,

maximum net import, and maximum net export. The same 1,649 contingencies are applied at every time in N1-3787. The panel contains 9,894 cases, of which 9,888 converge under the principal solution. Absolute screening passes 6,048 cases; 9,064 cases have neither a new violation relative to N=0 nor a material island. The latter metric does not detect every worsening of a pre-existing violation and does not replace absolute criteria.

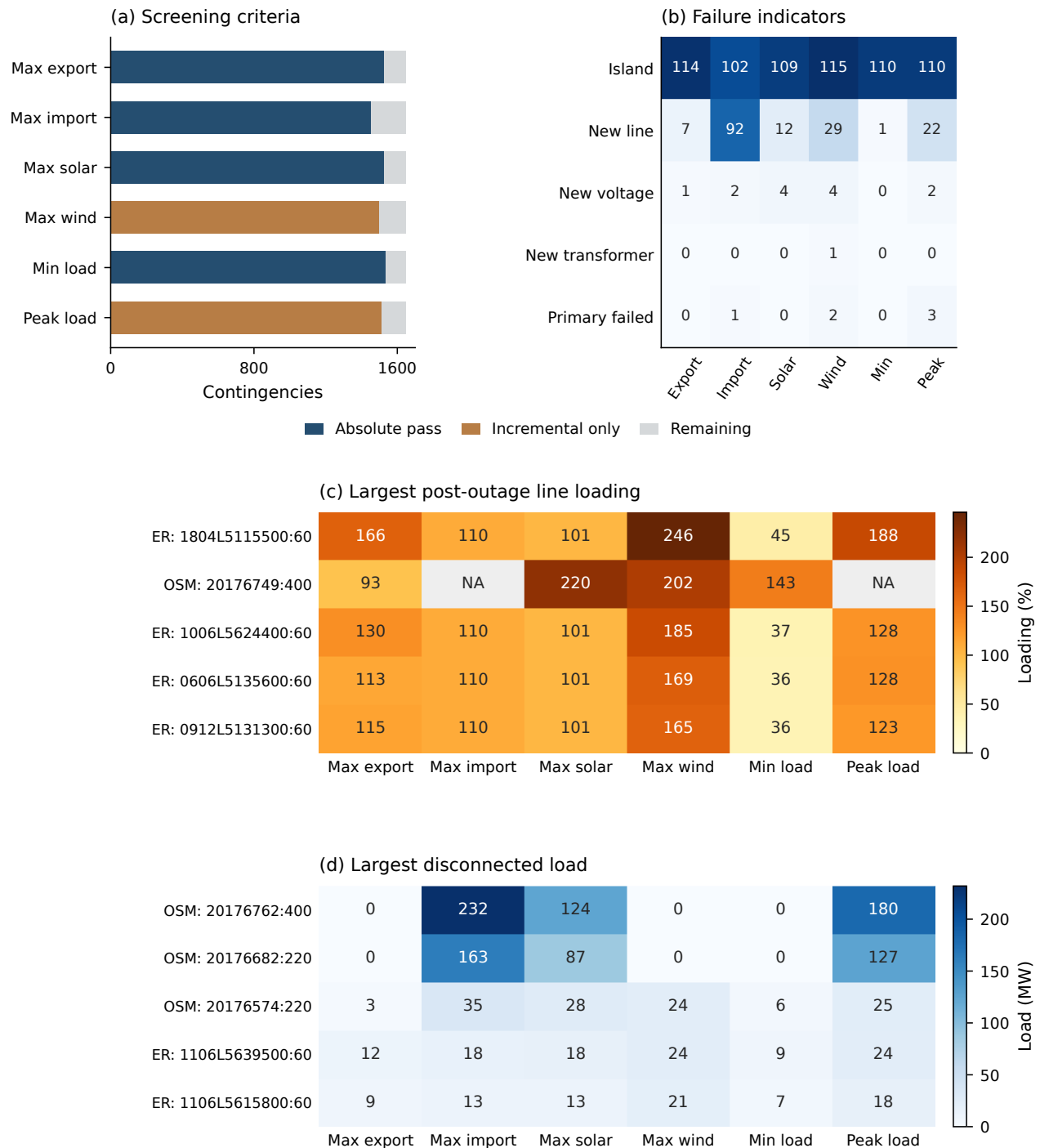


Figure S4 | Full-element N=1 screening at representative times. Absolute and incremental diagnostics, material islands, new violations, principal-solution failures, and the elements associated with the largest loading and disconnected load.

Table S10 | Full-element N–1 results at six representative times.

Snapshot	Cases	Principal solution converged	Absolute screen passed	No new violation and no material island	Material island
Maximum net export	1,649	1,649	1,527	1,527	114
Maximum net import	1,649	1,648	1,455	1,455	102
Maximum photovoltaic	1,649	1,649	1,526	1,526	109
Maximum wind	1,649	1,647	0	1,502	115
Minimum load	1,649	1,649	1,538	1,538	110
Maximum load	1,649	1,646	2	1,516	110
Total	9,894	9,888	6,048	9,064	660

The last three diagnostic columns are computed separately and are not mutually exclusive exhaustive classes. At maximum net export, eight additional non-islanding cases have new violations: seven line-thermal and one voltage violation. Together with 1,527 no-new-violation cases and 114 material islands, they account for all 1,649 cases.